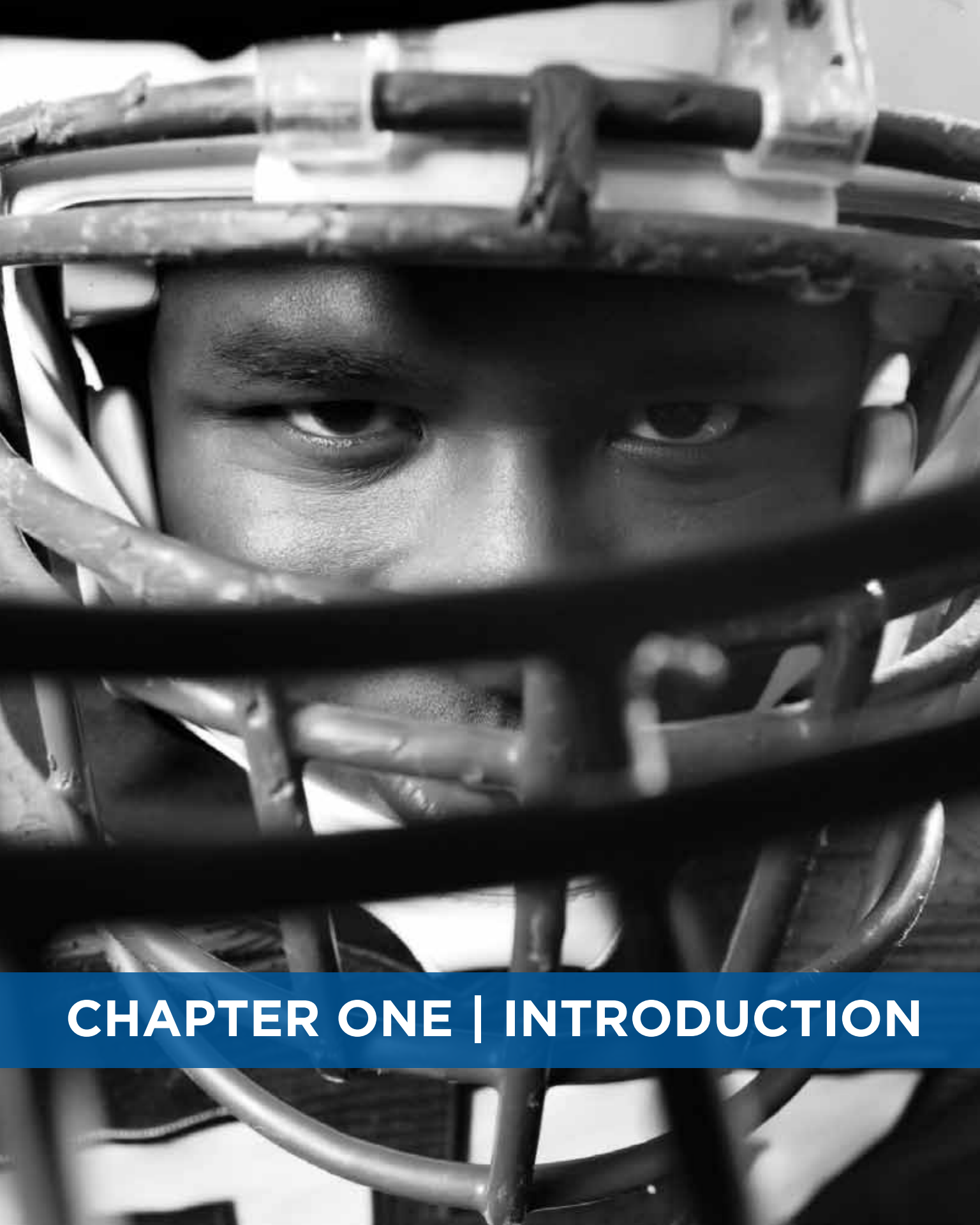




CHAPTER ONE | INTRODUCTION

Welcome to the National Strength and Conditioning Association's Basics of Strength and Conditioning Manual. The NSCA is the worldwide authority on strength and conditioning and this manual was developed to help you start your journey into the area of strength training and conditioning with some of the foremost coaches in the world. This manual is not meant to make you an expert, but rather to increase your knowledge, skills, and abilities with three goals in mind:

- Increase safety awareness
- Develop your abilities to supervise strength training and conditioning activities
- Provide an overview of the basic information needed to be effective as a strength and conditioning professional

What is Expertise?

The time required to be an expert in any given area has been estimated at 10,000 hr, or approximately 10 years, of direct practice (3). This manual is for those interested in strength training and conditioning but are still in the formative stages, in other words – you are not an expert. Moreover, it is unlikely you will ever know everything there is to know about strength training and conditioning. A look at the number of variables involved in strength training and conditioning results in a list of about 50 (e.g., sets, reps, weight, exercise selection, technique, time of day, temperature, training status, etc.). We calculated the number of possible combinations of these variables and came up with a number so large that no one could possibly study and know all the combinations in a lifetime of effort. This means that strength training and conditioning involves such a vast area of knowledge that much of our job will be to reduce the number of things to know to a manageable level by emphasizing those that are the most important. The important things are the “big ideas” or “big things.” It is vitally important that you get the “big things” right. The three objectives above, we believe, are the big things required for a basic understanding of strength training and conditioning.

Increase Safety Awareness

Clearly, first we must commit to doing no harm as strength training and conditioning professionals. Like all athletic activities, injury is a possibility and we must prepare such that we reduce the likelihood of injury. We will cover safety, injury prevention, and risk management in the final chapter to ensure you leave this manual with safety foremost in your mind. Fortunately, injuries in strength training and conditioning are rare but constant vigilance and good judgment are always required (6,7,8). Risk management is a tactic that is used to reduce the likelihood of injuries along with the likelihood of legal problems that often accompany injuries. Increasing your knowledge and awareness of the risks of injury through strength training and conditioning activities, and the risk of litigation or lawsuits due to poor judgment, we hope will keep athletes healthy and happy through competent strength training and conditioning decisions. We believe that the first step to safe performance is thorough and competent training of instructors and coaches.

Develop Your Abilities to Supervise Strength Training and Conditioning Activities

This manual draws a line between those who can design, administer, program, and plan strength training and conditioning activities, from those who can supervise and implement a program or plan. The knowledge, skills, and abilities needed to design a strength training and conditioning program require a higher level of knowledge than is covered in this manual. This manual will prepare you with a small amount of scientific information so that you can understand the basics of how strength training and conditioning affects the body, answer basic questions about training, and increase your scientific knowledge about training. The primary objective of this manual is to prepare you to identify flaws in exercise performance (e.g., posture and technique), fatigue, using too much resistance (or not enough), proper use of the appropriate metabolic energy system, and athlete readiness. As your knowledge and experience grow, you will acquire a “coach’s eye” which allows you to see flaws in exercise performance almost instinctively. You will also develop a sort of “sixth sense” regarding the status of your athletes such that you can tell when they are fatigued or lack the safety-related exercise preparation skills.

An Overview of Strength Training and Conditioning

Basic principles permeate all of strength training and conditioning (refer to the pyramid diagram for an idea of how this manual will attempt to “divide and conquer” the important basic information you will need to effectively supervise and direct day-to-day training). For example, one of the pillars of strength training and conditioning is the idea of progression. Progression refers to the selection of exercises, loads or resistances, order of exercises, and readiness of the athlete that are just right (not too hard, not too easy) for the status of the athlete and the demands of the activity. Another basic principle is that of specificity, the body tends to adapt very narrowly to the nature of the exercise performed. Finally, even supervision itself comes in different forms, and it is important to know the circumstances when a particular form of supervision is ideal. Continuing the idea of a broad overview, let us look at the basic principles of training and a few definitions (2,4,9).

Training – Training is the process of preparing an athlete physically, technically, tactically, psychologically, and theoretically rapidly for the highest levels of performance (4). Training involves more than simple growth and maturation and, of course, the highest levels of performance will be relative to the current status and genetic gifts of the athlete.

Volume – Volume is the amount of work performed. Sets and repetitions of an exercise combine to make volume (1). If you are a runner, volume is the distance you covered. If you are strength training, volume is the product of sets x repetitions of an exercise.

Intensity – Intensity is the difficulty of the work. Intensity is the amount of weight or resistance used in a particular exercise (1). If you are a runner, intensity is running speed. If you are strength training, intensity is the resistance or weight lifted.

Volume-Load – Volume-load is the combination of volume and intensity. Volume-load is usually calculated as sets x repetitions x weight, or resistance used (1).

Frequency – Frequency is simply the number of training sessions expressed per day, per week, per month, and so forth (1).

Principles of Training

1. Principle of Individuality

Every individual is unique and will respond differently to the same training stimulus. Some of these differences can be influenced by many characteristics; biological age, training age, gender, body size and shape, past injuries and many more.

For example, a college athlete makes a copy of his exact training program and gives it to his little brother who is a freshman in high school. The younger brother does not miss a workout, and at the end of the program, he is disappointed in the results. Though many variables could play a role in the results, the primary factor is most likely the large range in biological and training age.

2. Principle of Specificity

Training adaptations for an individual will occur specifically to the muscle groups trained, the intensity of the exercise, the metabolic demands of the exercise, and/or specific movements and activities. In an attempt to perfect a specific skill or activity, you must perform that skill or activity with proper body mechanics to have correct technique.

For example, a 100-m sprinter is not going to train for an event by running three miles at a low intensity for an extended period. The sprinter will train by sprinting short distances at very high intensities.

3. Principle of Overload

In order for an individual to achieve a certain training adaptation, the body must be stressed by working against a stimulus or load that is greater than that to which it is accustomed. Overload, ensures improvement by challenging changes in resistance, terrain, movement complexity, and many others.

For example, if an athlete is trying to increase force production to jump higher, the athlete must train to increase overall strength and power. When training with the hang clean at three sets of five repetitions, the athlete should load the bar with a weight that will allow them to use great technique at a desired velocity. If the bar is loaded with a weight that prevents them from reaching their desired velocity, then the specific training adaptation will not be obtained.

4. Principle of Progression

In order to achieve the desired training adaptations for a certain activity or skill consistently, the training stimulus must gradually and constantly increase. This implies that there is an optimal level and time frame for the overload to occur. If overload increases too quickly, poor technique, improper muscle firing patterns, and injury may result. If overload progresses too slowly, improvements will be minimal or non-existent. Rest and recovery must also be included in the progression, as training hard all the time could result in chronic fatigue, a decrease in performance and eventually injury.

For example, at the beginning of the training program, an athlete may be able to perform three sets of ten repetitions at 135 lb. At first, this may be a tough task for the athlete to accomplish, but as the athlete consistently trains, the task will become easier and the load must be increased. The next week the load increases to 145 lb until all ten reps can be completed with correct technique. The athlete must progressively overload the muscles to increase performance.

5. Principle of Diminishing Returns

Performance gains are related to the level of training (training age) of each individual. Athletes that have never participated in a training program before can see huge initial performance gains in their program. On the other hand, athletes that have been lifting for several years will see smaller gains over longer periods of time. As an athlete nears their genetic potential, the gains in performance will be much harder to obtain. The key is to continue to show progress in the areas in which they have weaknesses.

For example, when an athlete first starts a training program as a freshman in college the athlete's vertical jump may improve from 22 in. to 30 in. in the first year. As the athlete continues to train through the next three years, their vertical jump performance may increase from 30 in. to 38. Achieving the eight-inch improvement in the final three years is much more significant and difficult than the eight inches in the first year.

6. Principle of Reversibility

When a training stimulus is taken away from an athlete for an extended period of time, they will not be able to maintain a certain level of performance. Over time, the gains that were achieved will return to the original level.

For example, when an athlete takes the summer off from training they can expect to become detrained. The decrease in performance is directly related to the inactivity of the muscles that have been atrophied from nonuse (5).

These principles of training will guide your decisions and determine how you will accomplish the three objectives listed at the beginning of this chapter. These guidelines will help you judge the direction of different training approaches and likely outcomes. Moreover, training principles simply help reduce the universe of ideas of which you need to consider. Another aspect of strength training and conditioning is the utilization of proper energy systems.

Overview of Energy Systems

The following section will review the energy systems associated with the production and use of energy within the body. The most efficient and effective strength training and conditioning programs are designed with an understanding of the transfer of energy between all of the biological energy systems. It is beyond the scope of this manual to define the specific chemical reactions and processes within the body that generate and replenish energy. For more complete information about the energy systems, please refer to the “Essentials of Strength Training and Conditioning” (3rd ed.) textbook (1).

Through various reactions within the body, an intermediate molecule called adenosine triphosphate (ATP) is generated, which allows for the transfer of energy from various reactions. This molecule is classified as a high-energy molecule because it stores large amounts of energy in its chemical bonds. Through the breakdown of ATP, the body is able to provide the necessary energy to perform activities (without an adequate supply of ATP muscular activity and muscular growth would not be possible). Since ATP is stored in such limited quantities within muscle cells, three main energy systems exist in muscle cells to replenish ATP and provide the constant energy needed for muscle action (1). All of the energy systems are active all of the time; however, the contribution of one energy system may be dominant based on the intensity and duration of a specific activity.

Phosphagen (ATP-PC) System

The phosphagen, or ATP-PC, system uses ATP primarily for short-term, high-intensity activities (1). This energy system is considered anaerobic because it does not require the presence of oxygen to metabolize energy. However, this energy system does not generate ATP for immediate use and has a low capacity for ATP stores. Since the phosphagen system cannot store large amounts of energy, it is the dominant energy system for high-intensity activities that last no longer than 20 – 30 s (1). If an activity lasts longer than 20 – 30 s, energy supply shifts to the glycolytic energy system.

Glycolytic System

The glycolytic system has a much higher capacity for storing energy. This energy system, which is often broken into fast and slow glycolysis, is also considered anaerobic because it breaks down carbohydrates to produce ATP. Since it has a higher capacity for storing energy, this system is dominant during activities that last between 30 s and 2 – 3 min (1).

Oxidative System

The oxidative energy system is considered aerobic. Whereas the phosphagen and glycolytic energy systems derive energy from carbohydrates for high-intensity exercise, the oxidative energy system shifts from carbohydrates to proteins and fats as energy substrates for prolonged, submaximal exercise. The oxidative energy system is dominant for activities of low intensity that last longer than three minutes.

The body shifts between the various energy systems on a moment-to-moment basis in order to provide the energy necessary for movement and restoration. The energy systems respond to training specifically, as described by the principle of specificity. Since all energy systems are active all of the time, it is important to understand and differentiate when each system is dominant for each specific activity in order to design efficient and productive training programs that meet the needs of the activity. For example, a football lineman should not perform long-distance runs to train for competition because the demands of training do not match the demands of the activity.

Conclusion

While you are confronting the large number of demands that you will face as a relatively new strength training and conditioning professional, keep in mind that your primary tasks are safety, vigilant supervision, and a watchful eye of the implementation of the designed training program. The watchful eye comes from using the principles of training as a sort of criteria gauntlet through which any training decision should pass. Moreover, you can use the information in this chapter as a frame or context within which the following chapters fit.

Energy Systems

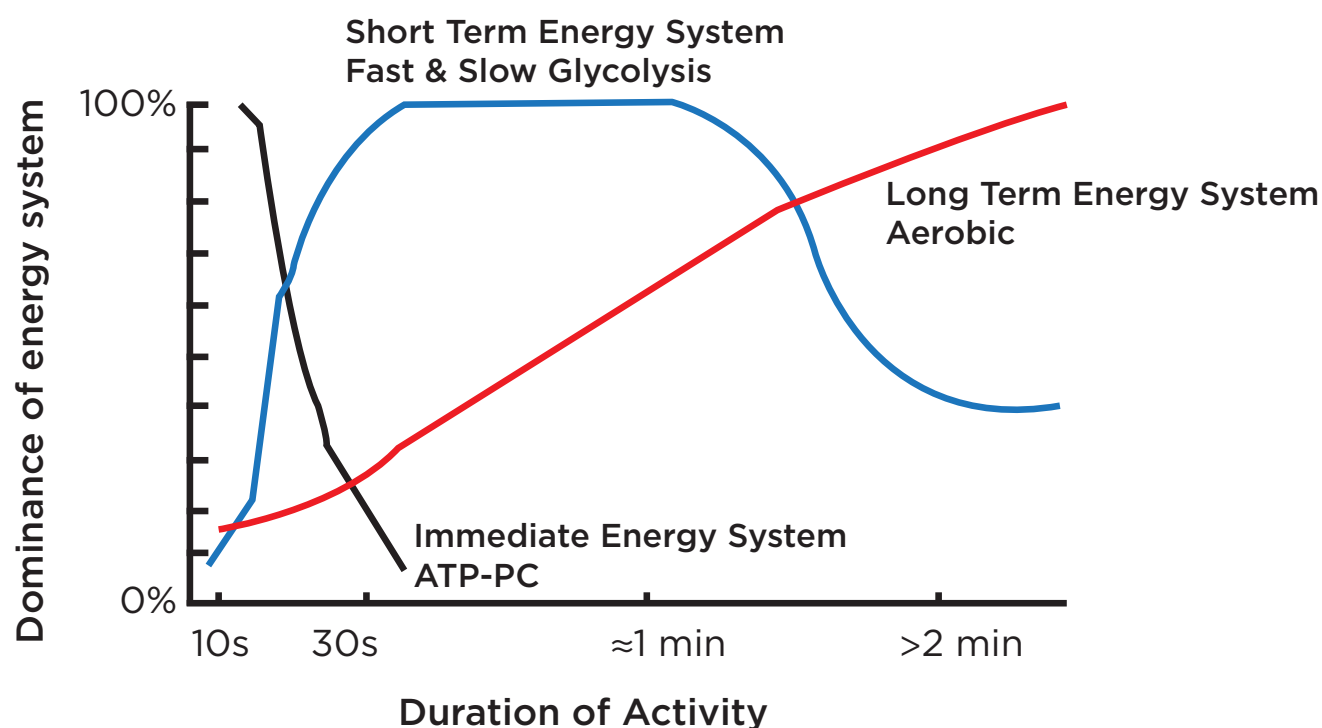
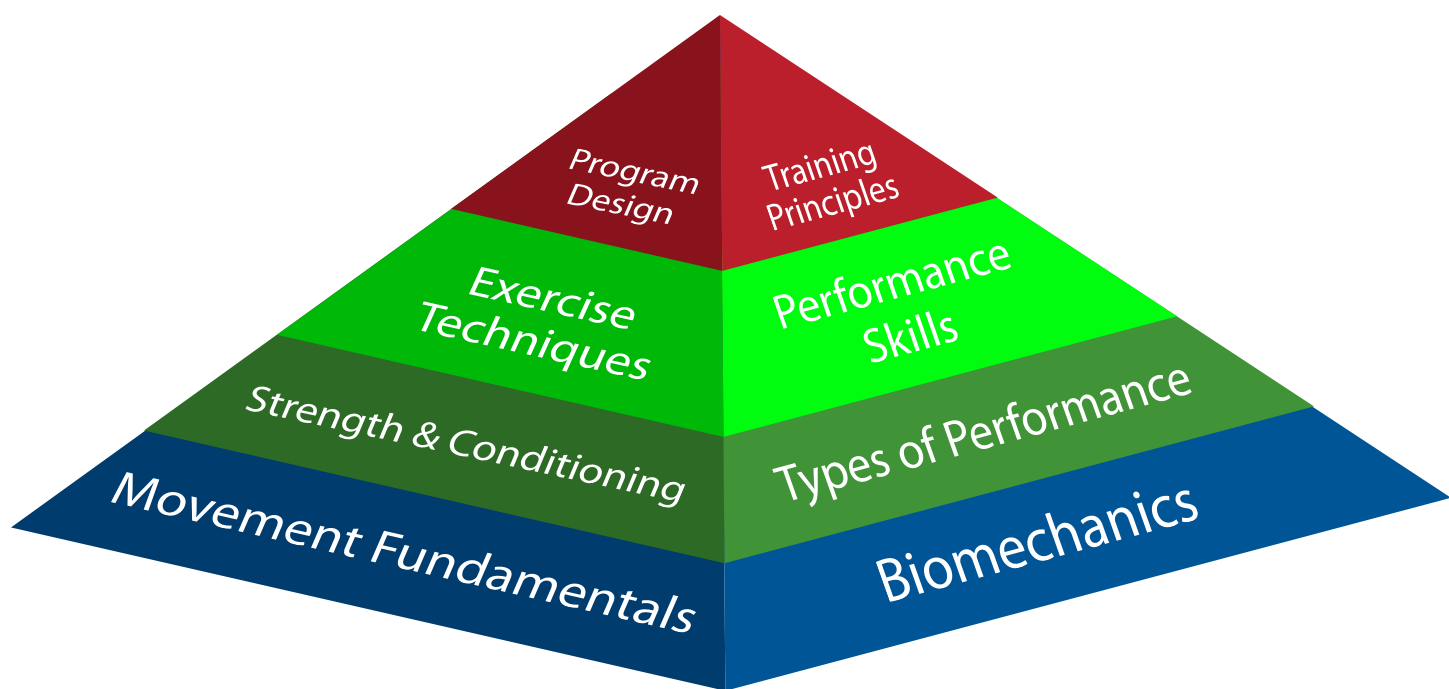


Figure 1-1. Dominant Energy System Based on Activity Duration

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Performance Pyramid